

Soohyun Cha (차수현)

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WORK EXPERIENCES

Samsung Research, Samsung Electronics

Seoul, Republic of Korea

Research Engineer at the AI Center

Mar. 2026 – **Current**

- (2026.07~) Efficient serving of LLMs on edge GPUs

Undergraduate Intern at the Communication Research Center

July 2022 – Aug. 2022

- Designing a deep learning-based KPI predictor for communication systems

EDUCATION

Seoul National University

Seoul, Republic of Korea

Master of Science in Electrical and Computer Engineering

Mar. 2024 – Feb. 2026

- Advisor: Prof. Jaewoong Sim
- Research interest: Computer systems and algorithms for efficient serving of emerging machine learning workloads, such as LLMs
- Thesis: Efficient Serving of Large Language Models with Adaptive Speculative Decoding
- GPA: 4.04 / 4.30

Bachelor of Science in Electrical and Computer Engineering

Mar. 2018 – Feb. 2024

- GPA: 3.74 / 4.30, Major GPA: 3.80 / 4.30 (Graduated with *Cum Laude*)
- Full-tuition scholarship for 8 semesters by the National Science and Engineering Scholarship, Korea Student Aid Foundation (KOSAF)

PUBLICATIONS

AdaTree: Serving-Aware Adaptive Tree Construction for Speculative Decoding

Soohyun Cha, Jungi Lee, Jaewoong Sim

The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS), Sydney, Australia, Dec 2026 (*To appear*)

MX+: Pushing the Limits of Microscaling Formats for Efficient Large Language Model Serving

Jungi Lee, Junyong Park, Soohyun Cha, Jaehoon Cho, Jaewoong Sim

Proc. of the 58th International Symposium on Microarchitecture (MICRO), Seoul, Korea, Oct 2025

TEACHING EXPERIENCE

Graduate Teaching Assistant at Seoul National University

- Digital Systems Design and Experiments, ECE 315.A Fall 2024
 - Lab sessions for Verilog and FPGA board practices
 - Project session for implementing end-to-end CNN accelerator on FPGA board
- Computer Organization, ECE 322 Spring 2024
 - Lab session for implementing a pipelined CPU in RTL and a cache simulator in C++

Undergraduate Course Tutor at Seoul National University

- Digital Systems Design and Experiments, ECE 315.A Fall 2023
- Basic Computing, Faculty of Liberal Education Fall/Spring 2023, Fall/Spring 2022
 - Outstanding tutor award (Fall 2023)

SKILLS

Programming Languages: Python, Verilog/SystemVerilog, C/C++, CUDA

Frameworks: PyTorch, Hugging Face Transformers, Triton, vLLM, SGLang

Tools: Nsight Systems, Xilinx Vivado

Platforms: Linux, Artix-7 FPGA board

Foreign Language: English (TOEIC Speaking: Intermediate High)

MISCELLANEOUS

Military Service in the Republic of Korea Army

July 2020 – Jan. 2022

Completed 18 months of mandatory military service in the Republic of Korea Army